

Exercise – 2 :- E-commerce Platform Search Function

Big O Notation

Big O notation is a mathematical representation used to describe the performance of an algorithm as the input size increases.

Common complexities:

Complexity	Meaning
$O(1)$	Constant Time
$O(\log n)$	Logarithmic Time
$O(n)$	Linear Time
$O(n^2)$	Quadratic Time

Search Operation Scenarios

Best Case

The target element is found immediately.

- Linear Search: First element matches.
- Binary Search: Middle element matches.

Time Complexity: **$O(1)$**

Average Case

The target element is located somewhere in the middle of the dataset.

- Linear Search: $O(n)$
- Binary Search: $O(\log n)$

Worst Case

The target element is the last element or is not present in the dataset.

- Linear Search: $O(n)$
- Binary Search: $O(\log n)$

Algorithm	Best Case	Average Case	Worst Case
Linear Search	$O(1)$	$O(n)$	$O(n)$
Binary Search	$O(1)$	$O(\log n)$	$O(\log n)$

Comparison

Linear Search

Advantages:

- Easy to implement.
- Works on both sorted and unsorted arrays.

Disadvantages:

- Searches every element in the worst case.
- Less efficient for large datasets.

Binary Search

Advantages:

- Faster searching.
- Efficient for large datasets.

Disadvantages:

- Requires sorted data.

For an e-commerce platform containing a large number of products, **Binary Search** is more suitable because it searches in **$O(\log n)$** time, making product retrieval much faster than Linear Search.